



Intelligent Rental Matching Web App

*Transforming Kenya's Rental
Experience*

Prepared By Group 1

Contents

- 01 Introduction: Revolutionizing Rental in Kenya
- 02 How Home-Match Works
- 03 Modeling Approach: Precision Matching Tech
- 04 Business Value
- 05 Deployment Plan
- 06 Future Enhancements
- 07 Conclusion



PART 01

Introduction: Revolutionizing Rental in Kenya

The Rental Landscape in Kenya



The Problem: Inefficiencies in the rental market

Renters face challenges with outdated or unavailable listings, leading to wasted time; property owners struggle to quickly find suitable tenants, resulting in extended vacancy periods and lost revenue.



The Opportunity: Untapped potential in a growing market

Kenya's rapid urbanization and the high demand for housing create a significant opportunity for PropTech solutions; intelligent solutions, leveraging real-time data, can streamline the rental process for both renters and owners.



Introducing HomeMatch: A smart PropTech solution

HomeMatch is designed to bridge the gap between renters and property owners by providing an intelligent, user-friendly platform; we aim to reduce inefficiencies, lower vacancy rates, and build trust in the rental market.

HomeMatch: Intelligent Matching Explained



HomeMatch Overview

HomeMatch connects renters and owners via a web-based dashboard using advanced matching, ensuring accessibility for all users.



Core Features Highlight

The platform provides geospatial search, fallback recommendations, and verified profiles to improve the rental experience.



Intuitive Interface

HomeMatch uses Streamlit for an interactive, user-friendly interface, ensuring quick and easy navigation.

HomeMatch: Seamlessly Connecting Renters and Owners

Advanced Matching Algorithms

01

HomeMatch uses advanced algorithms to connect renters and property owners effectively.

Web-Based Dashboard

02

The platform is a web-based dashboard, providing easy access and management for all users.

User-Friendly Interface

03

Ensures a positive experience with an accessible and easy-to-use interface.

Enhanced Rental Experience: Geospatial Search & Verified Profiles

Geospatial Search

Our platform offers geospatial search to improve the rental experience.



Verified User Profiles

Verified user profiles enhance trust and safety within the rental community.



Fallback Recommendations

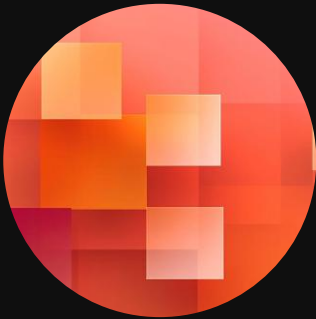
Fallback recommendations provide alternative options, ensuring users find suitable rentals.

HomeMatch: Streamlined User Experience with Streamlit



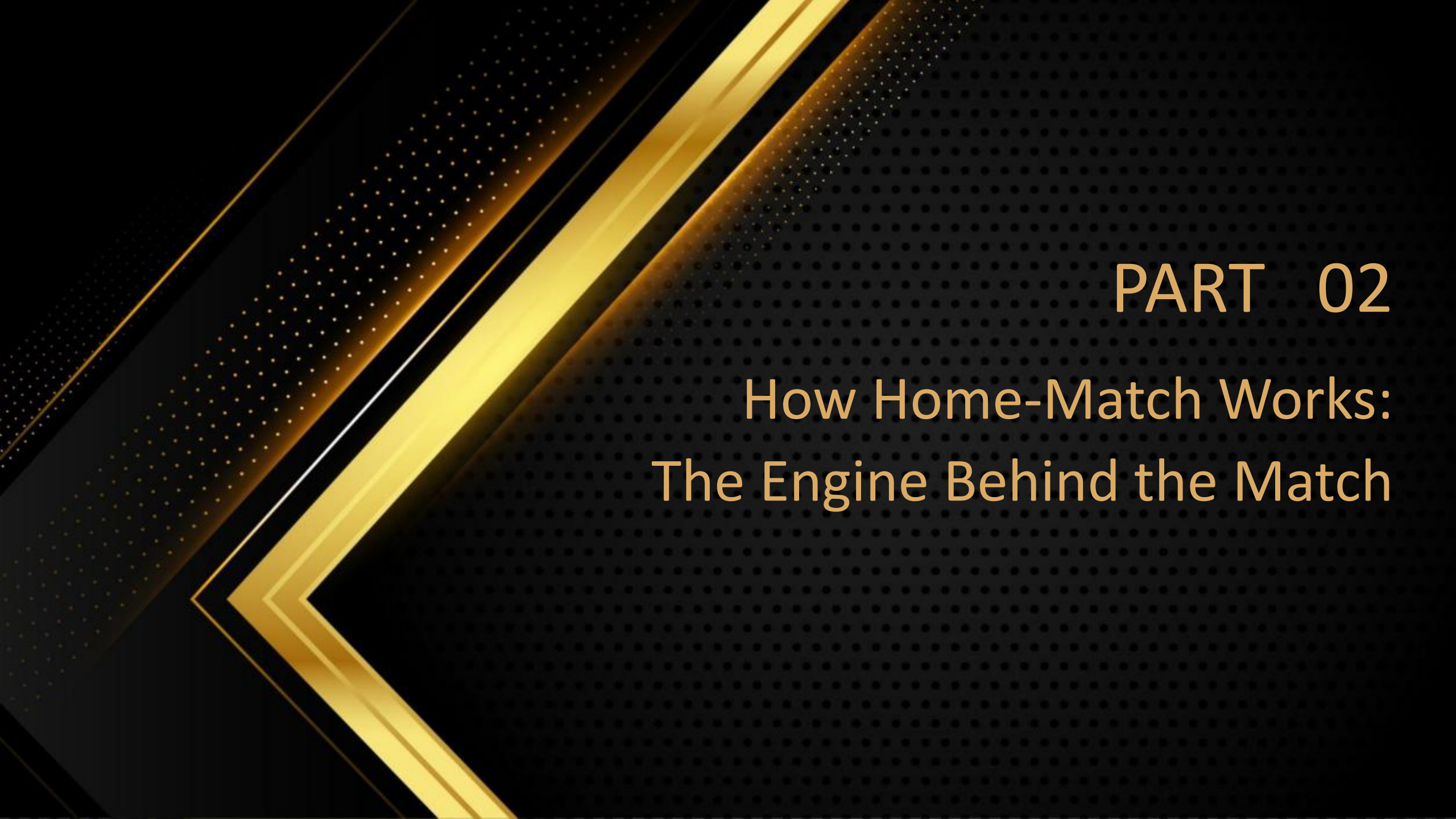
Interactive Interface Powered by Streamlit

HomeMatch utilizes Streamlit to deliver an interactive and user-friendly experience.



Intuitive Design for Easy Navigation

The platform's intuitive design allows users to easily navigate and find information quickly.



PART 02

How Home-Match Works: The Engine Behind the Match

Data-Driven Matching Process

Data Collection Overview

Describes the process of collecting and processing data from various sources to maintain a comprehensive property and renter database.

01

Recommendation Algorithms

Explains the use of algorithms in the recommendation engine for matching listings and providing real-time recommendations.

02

Dashboard Visualization

Details the Streamlit dashboard's interactive format for exploring properties and gaining insights into market trends and renter preferences.

03

Comprehensive Property Data Collection & Processing

Diverse Data Sources

Data is gathered from CSV files, APIs, public datasets, and manual inputs for a complete database.

Up-to-Date Information

Ensures the database contains current property listings and renter preferences.

Advanced Recommendation Engine: Real-Time, Relevant Listings

01

Advanced Matching Algorithms

Employs content-based similarity, geospatial nearest neighbors, and regression for accurate recommendations.

02

Real-Time Recommendations

Provides relevant listings and fallback recommendations in real time for optimal user experience.

03

Price Prediction via Regression

Utilizes regression techniques for price prediction, enhancing the relevance of listing recommendations.

Interactive Dashboard: Matches & Market Insights



Interactive Match Visualization

The Streamlit dashboard offers an interactive and visually appealing format for exploring property matches.

Insights for Renters & Owners

Renters can easily explore properties, while owners gain insights into market trends and renter preferences.

Data Overview: The Foundation of Our Matching

Listing Database Size

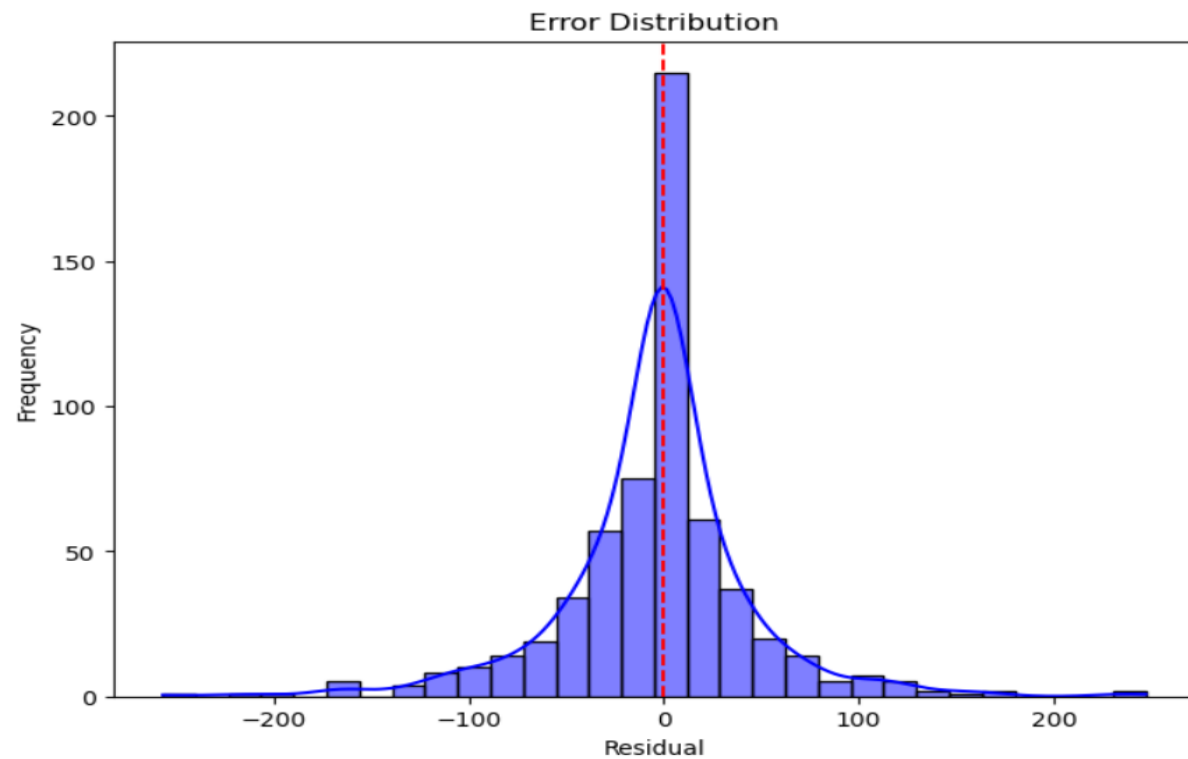
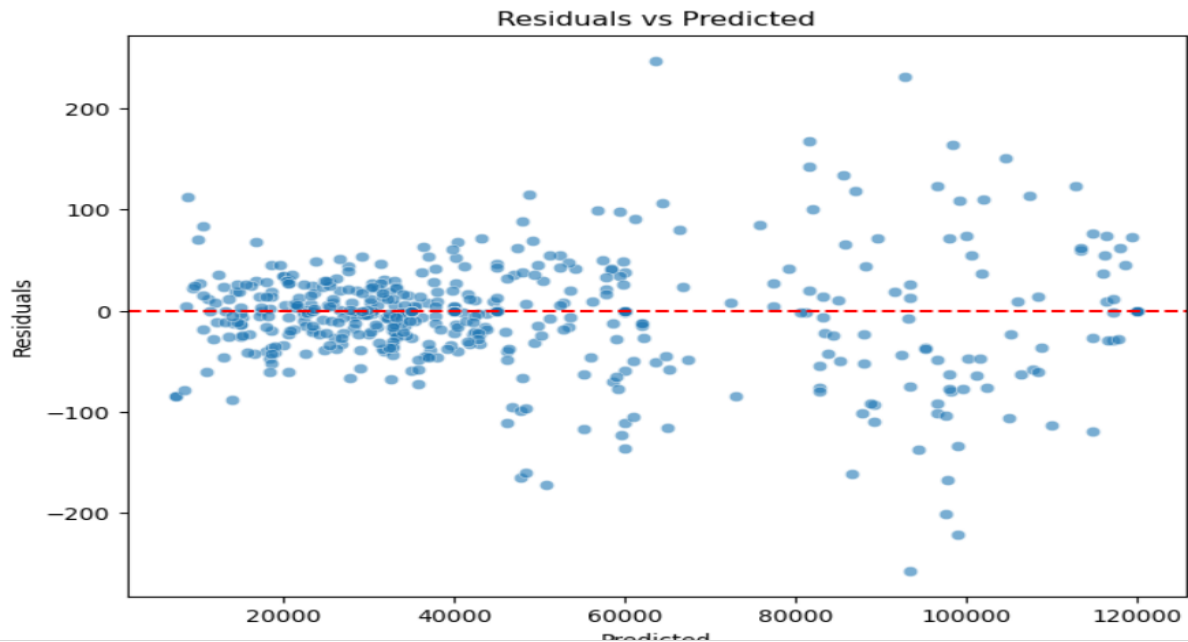
The platform has access to a large property database, containing 10,000-20,000 listings, which increases the chance of finding suitable matches for renters.

Key Listing Features

Listings include essential features like location, price, bedrooms, bathrooms and amenities, enabling accurate matching based on preferences and property traits.

Data Visualizations

The integrated dashboard uses price heatmaps, scatter plots, and vacancy maps to help renters and owners understand market dynamics for better decisions.



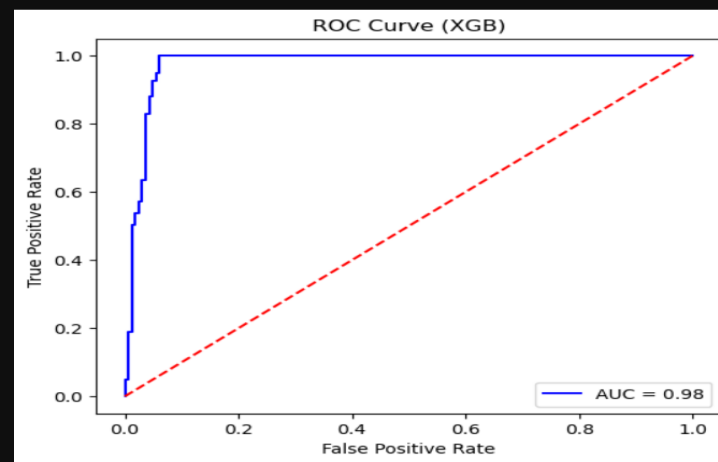
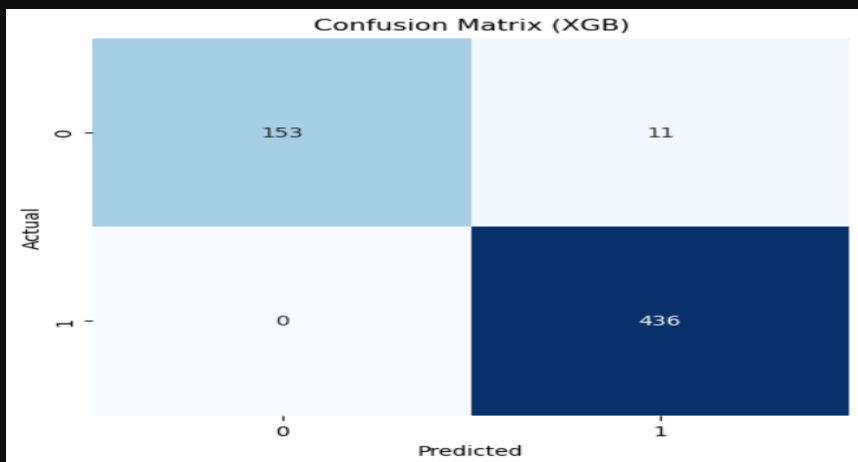
Extensive Property Listings: 10,000-20,000 Options

Our platform provides access to a vast database of properties, ensuring a diverse selection for renters and increasing the chances of finding the perfect match.

10,000–20,000
Property Listings

Extensive database offering a wide range of options for renters.

Key Features for Accurate Property Matching



Comprehensive Listing Features

Listings include location, price, bedrooms, bathrooms, and amenities for accurate matching.

Data-Driven Matching

Comprehensive data enables matching based on renter preferences and property characteristics.

Visualizations: Market & Property Insights

price heatmaps

Visualization Type

Used to provide market and property insights.

scatter plots

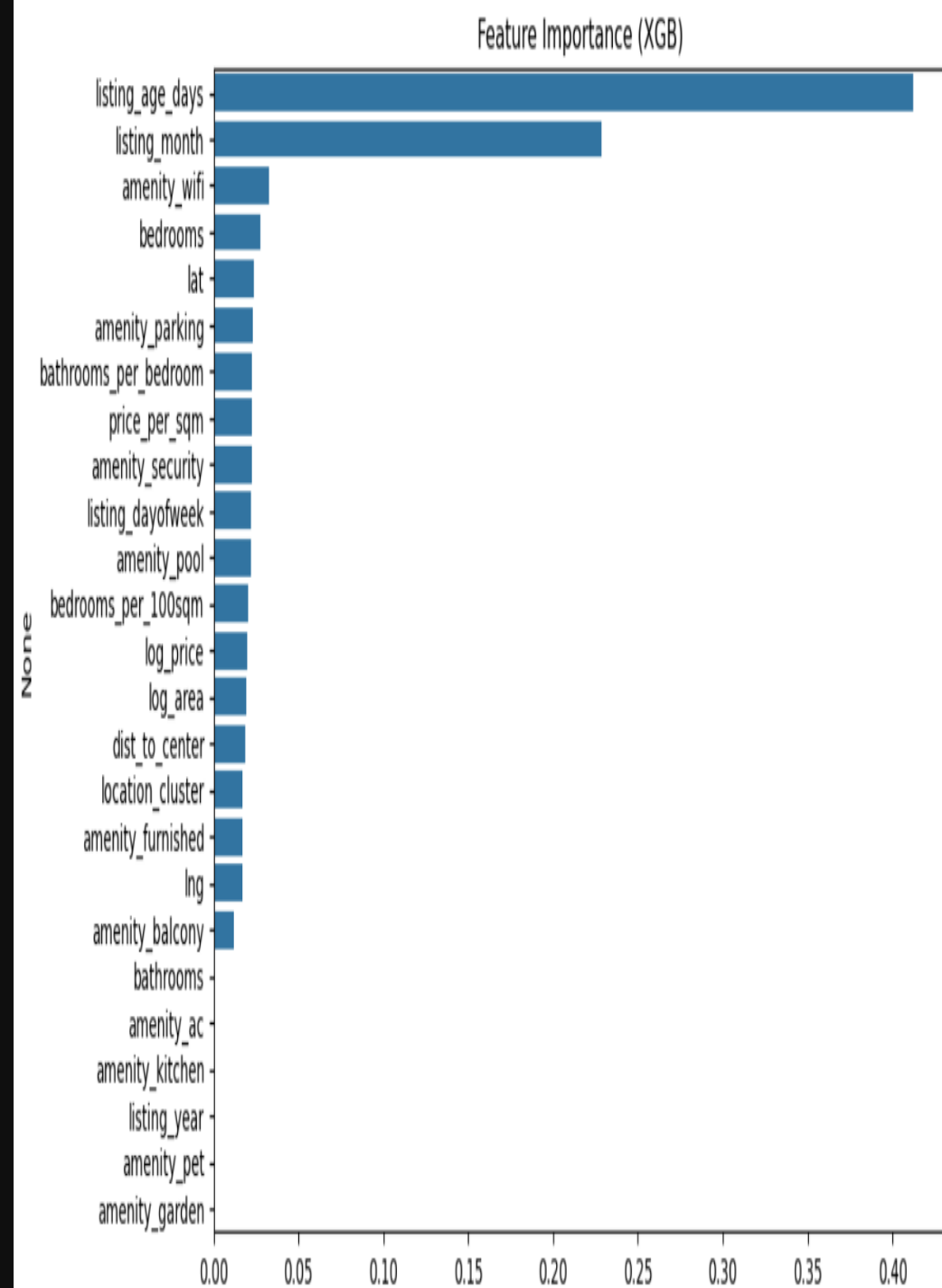
Visualization Type

Used to provide market and property insights.

vacancy maps

Visualization Type

Used to provide market and property insights.



The background features a dark, textured surface with a fine grid of small dots. Overlaid on this are several bright, glowing gold geometric shapes, including a large chevron pointing left and several parallel diagonal lines. The gold elements have a soft, hazy glow around them.

PART 03

Modeling Approach: Precision Matching Tech

Recommendation Algorithms: Matching renters with properties



Content-Based Recommendation: Matching based on property features

We use content-based recommendation, leveraging techniques like cosine similarity to match renters with properties based on features ; this ensures that renters see listings that align with their stated preferences.



Geospatial Search: Accurate location-based matching

Our platform incorporates geospatial nearest-neighbor search to ensure accurate location based matching; renters can easily find properties that are close to work, school, or other important locations.



Price Prediction: Ensuring fair market value

We employ a regression model for price prediction, which helps ensure that properties are listed at a fair market value; this enhances transparency and trust in the rental process.

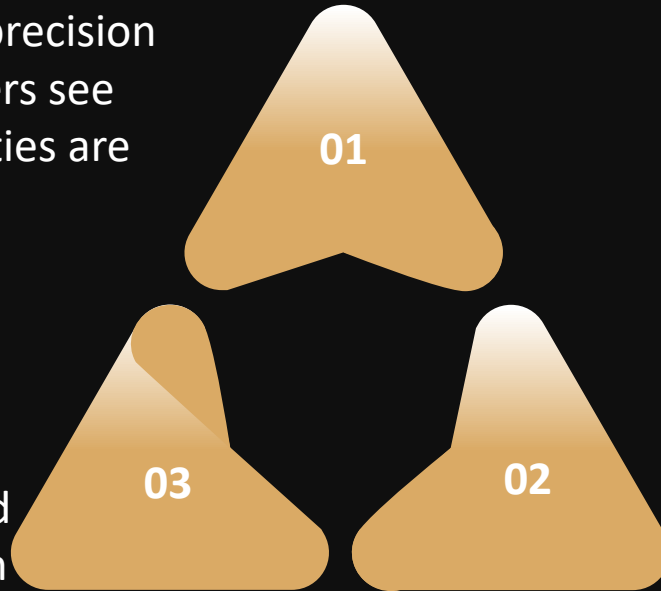
Evaluation Metrics: Measuring Performance

Precision and Recall: Ensuring accuracy of recommendations

We track Precision@K and Recall@K to evaluate recommendation quality; maintaining high precision and recall is essential for ensuring that renters see relevant listings and that no suitable properties are missed.

Business Focused Metrics: Optimizing for results

Home-Match optimizes for faster matches and higher satisfaction to ensure that the platform is providing value to both renters and owners; these business-focused metrics drive the platform's ongoing development and improvement.



User Engagement: Measuring user satisfaction

User engagement is measured through clicks, favorites, and matches, providing insights into overall user satisfaction; high user engagement is a key indicator of HomeMatch's success in meeting renter and owner needs.



PART 04

Business Value: Time Savings,
Lower Vacancy, and Trust

Benefits for Renters



Faster Searches: Quickly finding suitable properties

Renters experience faster searches, reducing the time and effort required to find a suitable property; optimized search and matching algorithms streamline the process, making HomeMatch a valuable tool.

Increased Choices: Access to a wide range of listings

Renters gain access to a wide range of listings, increasing their chances of finding the perfect home; a comprehensive database enhances the likelihood of finding suitable options that meet individual needs and preferences.

Benefits for Property Owners



Efficient Tenant Acquisition

Property owners benefit from lower vacancy rates due to intelligent matching and verified profiles, maximizing rental income.



Data-Driven Optimization

Owners use data insights to optimize property management strategies, understanding market trends and renter preferences.



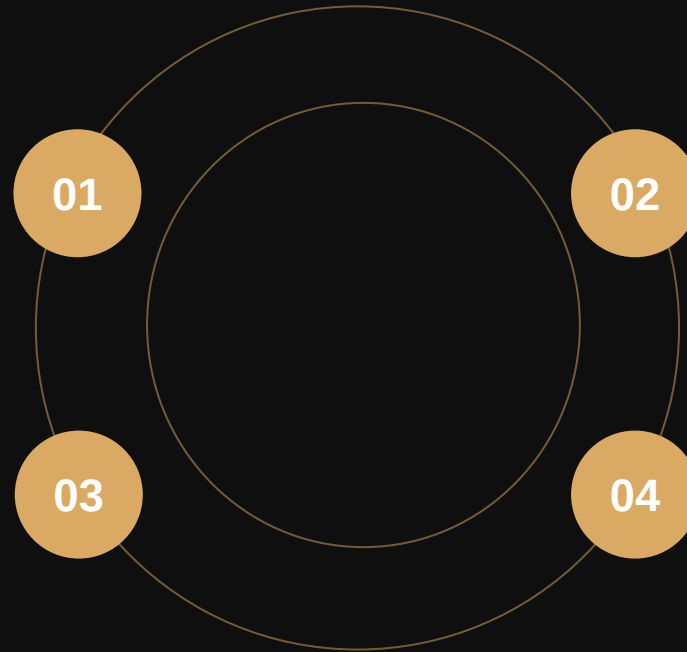
Maximize Rental Income: Lower Vacancy Rates with Efficient Tenant Acquisition

Reduced Vacancy Rates

Property owners experience lower vacancy rates through efficient tenant acquisition strategies.

Verified User Profiles

Verified user profiles ensure properties are matched with reliable and qualified tenants.



Intelligent Matching Algorithms

Intelligent matching algorithms quickly connect properties with suitable tenants.

Maximized Rental Income

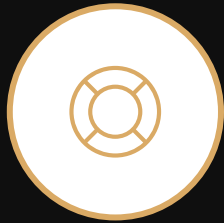
Efficient tenant acquisition maximizes rental income by minimizing vacancy periods.

Data-Driven Property Management: Optimize with Insights



Leverage Data-Driven Insights

Owners use platform data to optimize strategies.



Understand Market Trends

Gain insights into market trends for informed decisions.



Renter Preferences Analysis

Analyze renter preferences to improve property appeal.



Property Performance Tracking

Track property performance for better management.

Building Trust and Reliability

Verified Users: Ensuring the legitimacy of listings

Verified users help build trust and ensure the legitimacy of listings and renter profiles; our verification process helps to prevent fraudulent activities, creating a secure and reliable environment for all users.

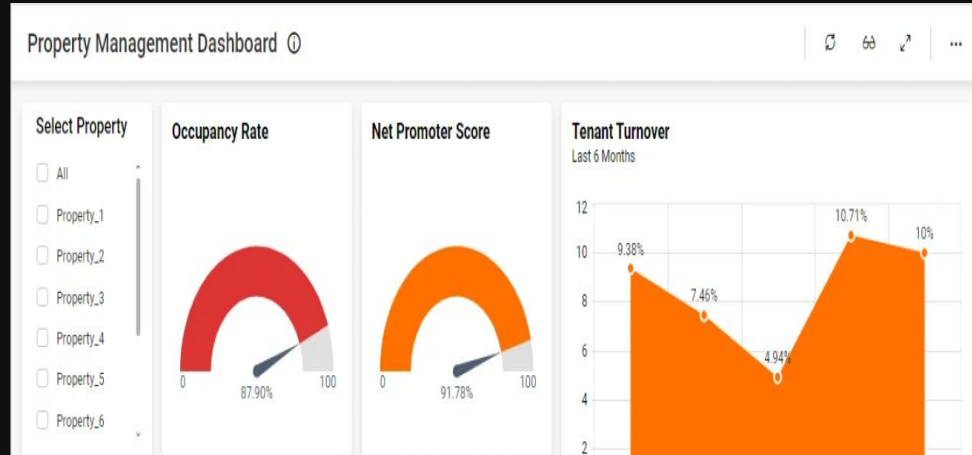


The background features a dark, textured surface with a fine grid of small dots. Overlaid on this are several bright, glowing gold geometric shapes, including a large chevron pointing left and several parallel diagonal lines that create a sense of depth and movement.

PART 05

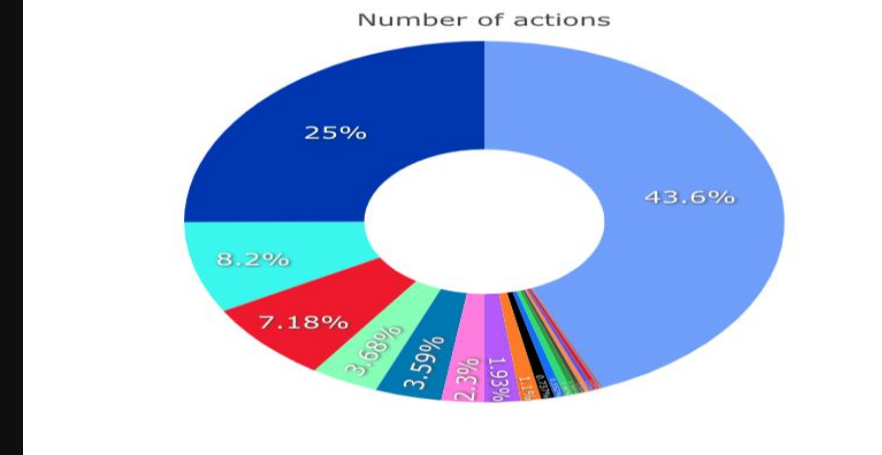
Deployment Plan: From MVP to Scalable System

Current Deployment: Streamlit and PostgreSQL



01 Streamlit Interface: User friendly presentation of data

We currently use Streamlit for our Minimum Viable Product (MVP) deployment; the user-friendly interface allows for easy exploration of data and matching results.



02 FastAPI Backend: Connecting data and presentation layers

Our backend is built using FastAPI and a PostgreSQL database to manage property listings; this robust architecture supports efficient data management and reliable platform performance.

Future Scaling: Tech stack for next level growth

React/Next.js Front End: Creating a dynamic user experience

Future scaling will involve transitioning to React/Next.js for the front end to create a more dynamic and interactive user experience; this will improve user engagement and overall satisfaction.

Cloud Hosting: Ensuring scalability and reliability

Cloud hosting will allow us to scale our infrastructure and ensure reliability; scaling the infrastructure using cloud services guarantees optimal performance during peak periods.





PART 06

Future Enhancements: Advanced Features On The Horizon

Extending the Platform

01

Mobile app: Expanding Accessibility

Mobile app integration is planned to enhance accessibility, enabling renters and owners to manage listings on the go; mobile app support makes HomeMatch more convenient for users who prefer to use their smartphones and tablets.

02

Fraud Detection: Ensuring secure experience

Fraud detection and ID verification will further enhance security and build trust; an improved security framework minimizes the risk of fraudulent accounts and activities.

Advanced Analytics: Gaining more insights

01

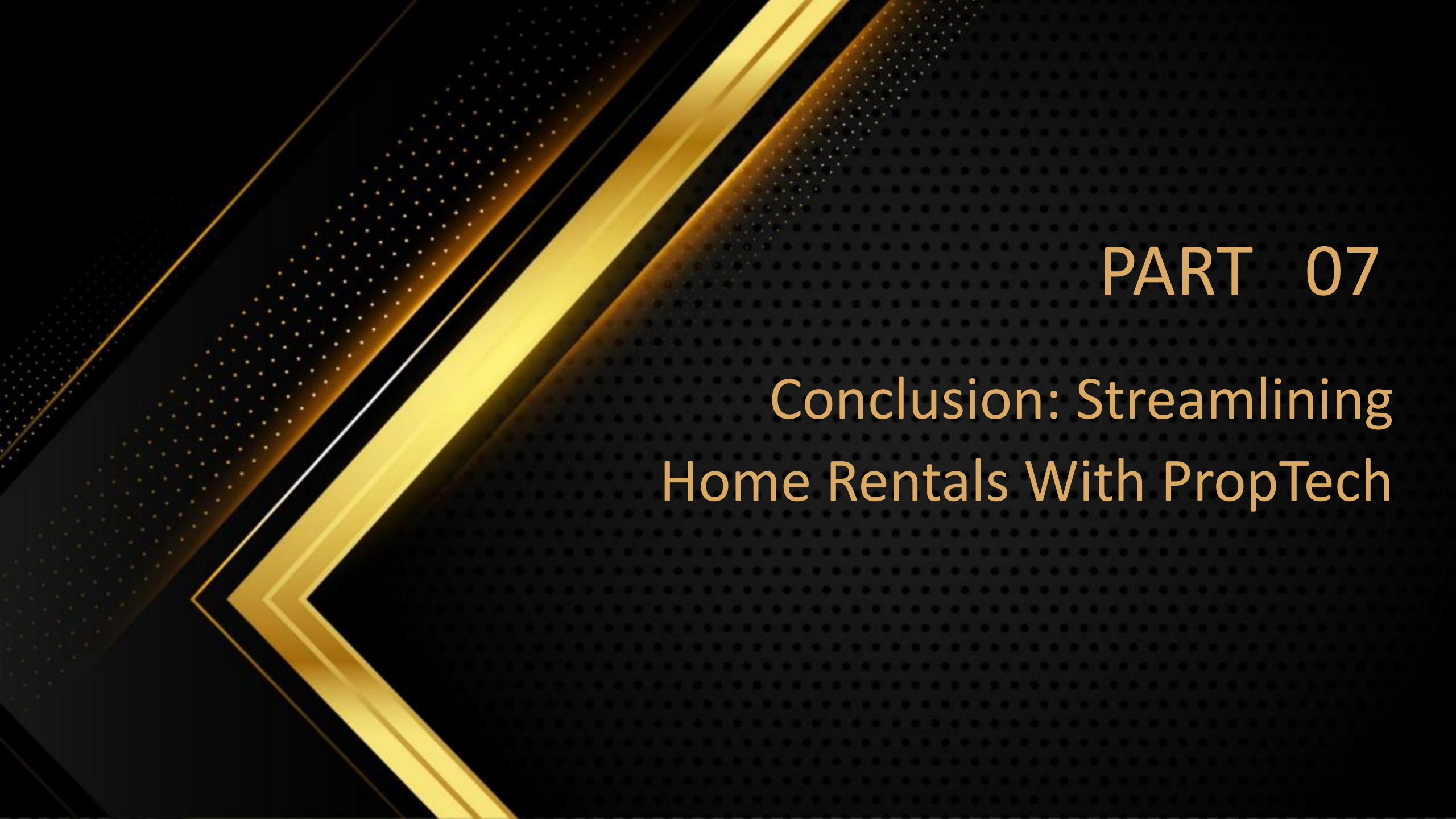
Real-Time Availability: Keeping the platform accurate

Real-time availability synchronization will reduce frustration caused by outdated listings; this feature assures that properties are accurately represented with up-to-date data.

02

Analytics Dashboards: Helping with decision-making

Advanced analytics dashboards offer detailed property management insights, helping owners make data-driven decisions; an improved analytics dashboard enhances strategic planning and property usage.

The background features a dark, textured surface with a grid of small, light-colored dots. Overlaid on this are several bright, glowing gold geometric shapes, including a large, multi-layered chevron pointing towards the bottom left and several parallel diagonal lines that create a sense of depth and movement.

PART 07

Conclusion: Streamlining Home Rentals With PropTech

HomeMatch: Redefining how properties are matched

01

Transforming the Rental Process: Simpler and Faster

HomeMatch streamlines Kenya's rental process and creates a simpler, faster, and more efficient experience for renters and property owners; the transformed system saves time, reduces vacancy, and makes the rental process more predictable.

02

Smart Matches, Simpler Homes: Efficient and effective

Our vision is to deliver smart matches and simpler homes for renters across Kenya using the features offered by the HomeMatch platform; an efficient and effective system ensures positive experiences.

Next Steps: Expanding reach



Pilot Launch Strategy

The pilot launch is designed to refine HomeMatch features using real-world user feedback from a control group.



User Testing Importance

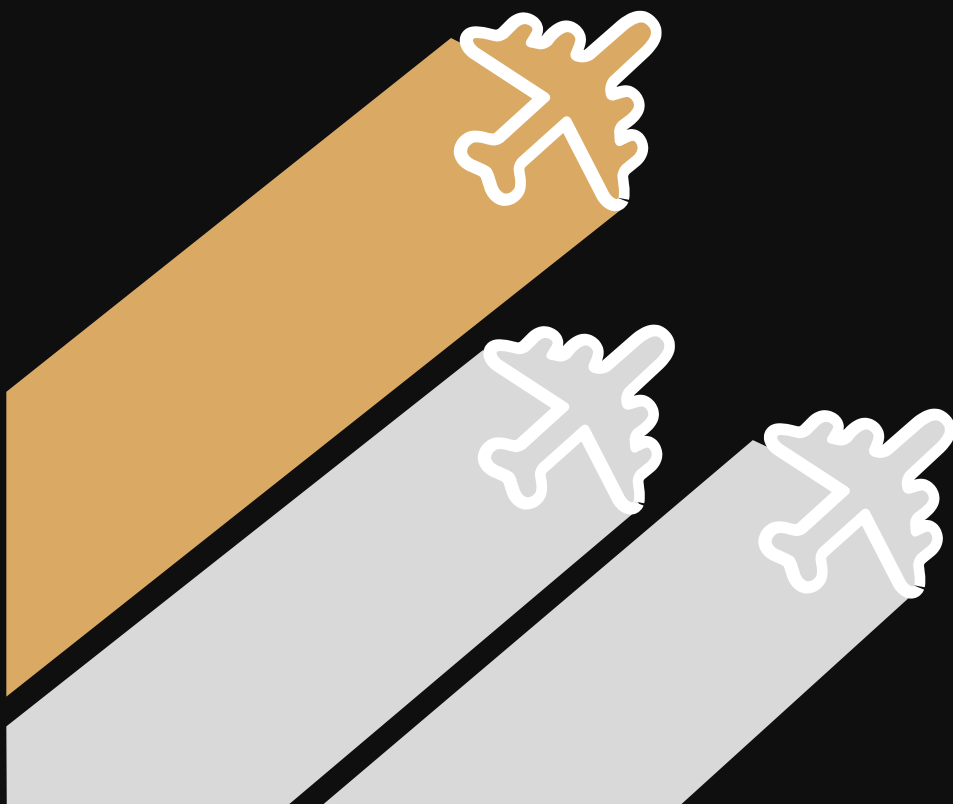
User input and testing enhance HomeMatch's user-friendliness and effectiveness, driving continuous product improvement.



Investor Outreach Goals

Investor outreach aims to secure funding for HomeMatch's expansion and increased value to Kenya's rental market.

HomeMatch Pilot Launch: Refine Features with User Feedback



Pilot Launch with Key Users

HomeMatch is launching a pilot program with key users and partners to gather real-world feedback.

Refining Features Based on Feedback

The pilot launch will allow for refining HomeMatch's features based on initial findings.

Actionable Insights from Control Group

Piloting within a control group generates actionable insights for improvement.

HomeMatch: Enhancing User Experience Through Testing

Valuing User Input

HomeMatch prioritizes user feedback from both renters and owners to ensure a user-friendly and effective platform.

Continuous Product Improvement

User testing is integral to HomeMatch's ongoing efforts to refine and improve its product based on real-world user experiences.

HomeMatch: Funding for Growth in Kenya's Rental Market

Seeking Investment for Expansion

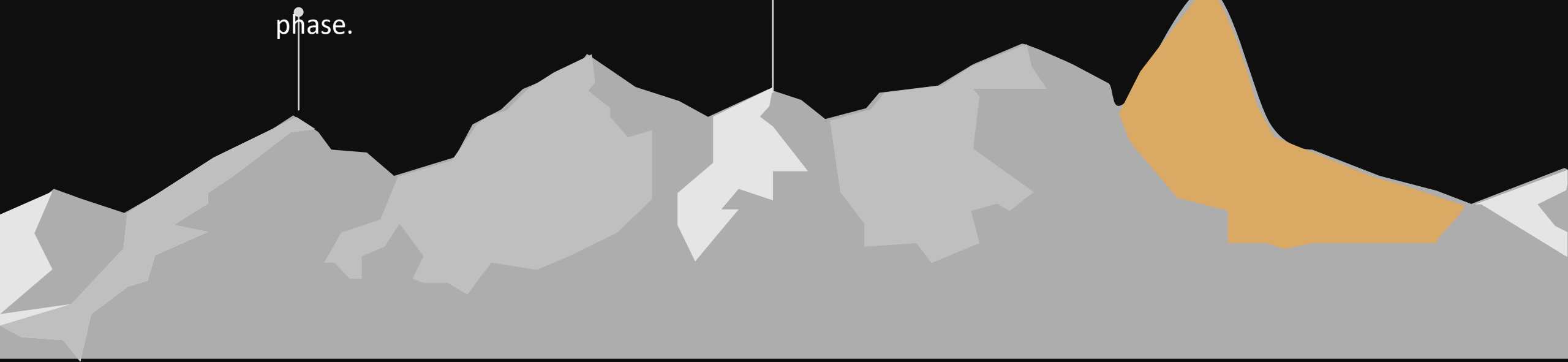
HomeMatch is seeking investor support to facilitate growth and scale its operations.

Extending Reach in Kenya

Funding will enable HomeMatch to extend its reach and provide increased value to Kenya's rental market.

Beyond Prototype Phase

Secured funding will allow HomeMatch to expand beyond its initial prototype phase.



THANK YOU



Prepared By Group I Collaborators

1. Anthony Muthee Njiru
2. Beth Mwangi
3. Edwin Joshua
4. Kelvin Shilisia
5. Mercy Chebet
6. Mirriam David